Week 3

Project Planning, Scope & Selecting the Project Approach

From an authorized project to a clear plan of what will be delivered and how it will be managed.



Where Are We in the Project Management Cycle?

Planning begins after authorization



Week 3 starts detailed planning
Scope • Approach • Requirements • WBS

From Week 2 to Week 3

The management question changes

Week 2 — Authorization

Should we undertake this project?
Is it feasible and valuable?
Which alternative is best?
What does the sponsor approve?

Week 3 — Planning

What exactly is included?
Which approach fits this project?
What products and activities are required?
How will scope be controlled?

Week 3 Big Question

How do we turn an approved idea into a manageable project?



Today we define: scope, objectives, approach, requirements, deliverables, WBS, and scope control.

Learning Outcomes

By the end of Week 3, students should be able to...

- Explain the Step Wise project-planning framework.
- Distinguish product scope, project scope, requirements, deliverables, and acceptance criteria.
- Select an appropriate project approach: predictive, iterative, Agile, or hybrid.
- Develop a project scope statement and identify exclusions.
- Construct a Work Breakdown Structure and WBS Dictionary.
- Recognize scope creep and explain how scope is controlled.

Lecture Roadmap

A practical planning sequence

Step Wise planning

Project approach

Scope & requirements

Deliverables & acceptance

WBS & WBS Dictionary

Scope control

Pattern: concept → example → interactive decision → application to student project

Step Wise: Planning Is More Than a Gantt Chart

Hughes, Cotterell & Mall — planning logic

Should we do it?

What are the objectives?

How is it organized?

What kind of project?

What products/activities?

How much effort?

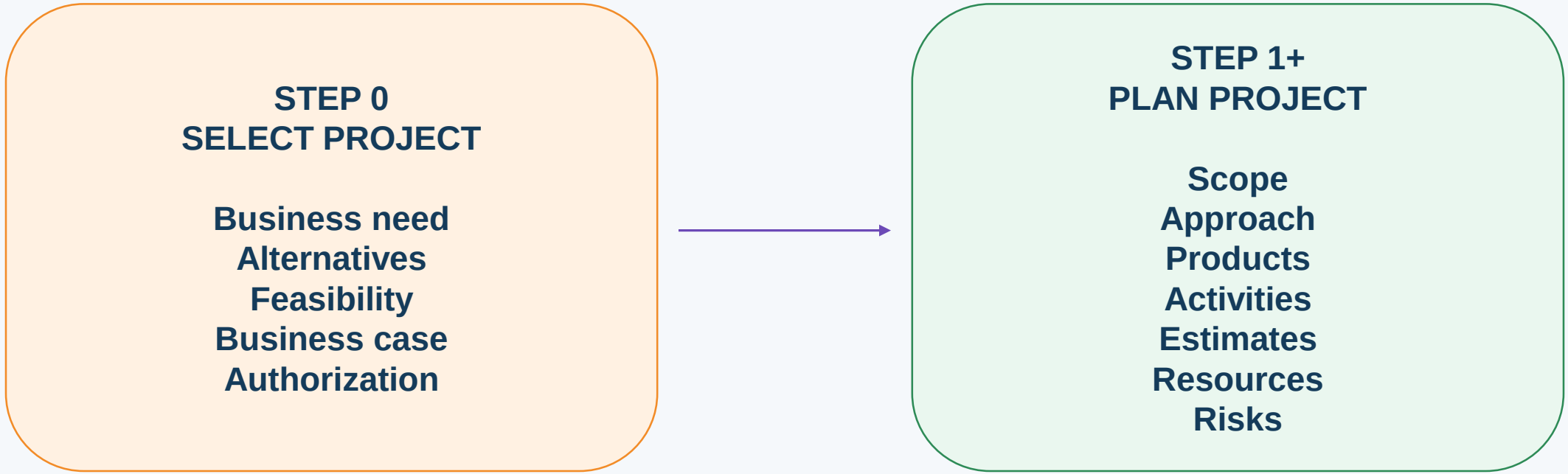
What risks?

Who does the work?

Is the plan realistic?

Step 0 Was Week 2: Select the Project

Now we start planning the selected project



Week 3 answer: What exactly are we planning and how will we plan it?

Step 1: Identify Scope and Objectives

The foundation of the plan

- **Objectives**
- **Authority**
- **Stakeholders**
- **Communication**
- **Scope boundaries**
- **Measures of success**

If objectives or boundaries are unclear, estimation and scheduling become unreliable.

Step 2: Identify Project Infrastructure

How the project will be supported and governed

Infrastructure Includes

- Project organization
- Reporting structure
- Development environment
- Tools and repositories
- Standards and templates
- Review/approval routes

ABC Retail Example

- Sponsor: Sales Director
- PM: student team lead
- Tools: MS Project + Jira + GitHub
- Reviews: weekly status report
- Approvals: sponsor sign-off

Step 3: Analyze Project Characteristics

The approach depends on the nature of the project

- **Novelty**
- **Complexity**
- **Uncertainty**
- **Criticality**
- **Team experience**
- **User involvement**
- **External dependencies**
- **Regulation/security**

A payroll system replacement, a research prototype, and a mobile startup app should not be managed in exactly the same way.

Step 4: Identify Products and Activities

What will be produced, then what work is needed



Week 3 focuses mainly on products/deliverables and decomposition. Week 5 converts work packages into schedule activities.

Step Wise and Our Course Map

Where the planning steps appear across weeks

Step Wise	Course week
Step 0: Select project	Week 2
Step 1: Scope & objectives	Week 3
Step 2–4: infrastructure, characteristics, products	Week 3
Step 5: Estimate effort	Week 4
Step 6: Identify risks	Week 7
Step 7–8: resources and plan review	Week 6 / Week 8
Step 9–10: execute and refine	Weeks 10–12

Selecting the Project Approach

A planning decision before estimation and scheduling

Predictive

Iterative

Agile

Hybrid

The selected approach affects requirements, estimation, scheduling, quality, monitoring, and change control.

Why Approach Selection Matters

The same project can be planned differently

Predictive mindset

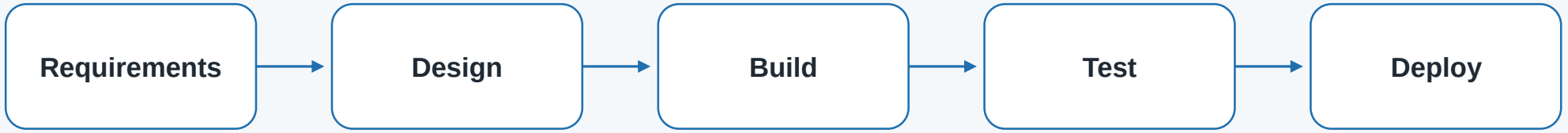
- Define most scope early
- Build detailed WBS and schedule
- Control changes formally
- Useful when requirements are stable

Agile mindset

- Prioritize backlog
- Deliver increments
- Adapt through feedback
- Useful when requirements evolve

Predictive Approach

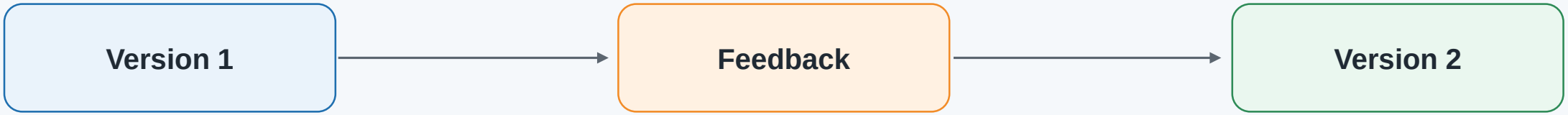
Plan-driven development and baseline control



Best when requirements are relatively stable, compliance is high, or formal approval and baselines are required.

Iterative and Incremental Approaches

Repeated learning and staged delivery

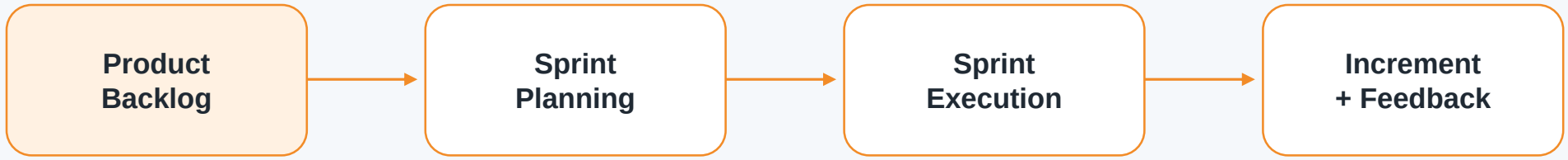


Incremental delivery: release parts of the product.

Iterative development: improve through repeated cycles.

Agile Approach

Manage uncertainty through short cycles and feedback



Useful when user needs are evolving and frequent feedback is valuable.

Hybrid Approach

Real projects often combine planning models

Management level
Milestones • budget • governance

Development level
Sprints • backlog • increments

Example: a fixed academic deadline, but iterative development of prototype features.

Approach Selection Matrix

A quick decision tool

Condition	Leans predictive	Leans Agile / iterative
Requirements	Stable, known early	Unclear or changing
Customer availability	Limited after approval	Available for feedback
Criticality/regulation	High formal control	Moderate / adaptable
Technology uncertainty	Low	High or experimental
Delivery value	One major release	Useful increments possible

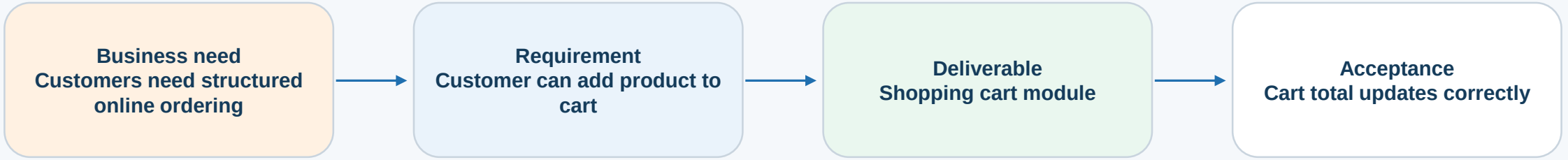
Interactive: Which Approach Fits?

Vote and justify your decision

A hospital must replace a reporting module by a legal deadline.	**Predictive**
A startup is exploring a new mobile social feature.	**Agile**
A university needs a fixed registration portal but wants student feedback on UI.	**Hybrid**

From Business Need to Requirements

Requirements translate need into system behavior



Good requirements are the bridge between project authorization and scope planning.

Types of Requirements

Different requirements affect planning differently

Functional
What the system does

Non-functional
How well it performs

Interface
How it connects

Data
What data is needed

Security
Who can access what

Reporting
What information is produced

Product Scope vs Project Scope

Two different but connected questions

Product Scope

- Features and functions of the software
- Example: login, search, cart, reports
- Focus: what the product will contain

Project Scope

Work required to deliver the product

Example: analysis, design, build, test, training

Focus: what the project team must do

Scope Statement: What It Should Contain

A management agreement, not a wish list

1. Project purpose and objectives
2. High-level product scope
3. Major deliverables
4. Acceptance criteria
5. In-scope and out-of-scope items
6. Assumptions and constraints
7. Major boundaries and exclusions

In Scope vs Out of Scope

Boundaries prevent uncontrolled growth

In Scope — first release

- Customer registration
- Product catalog
- Shopping cart
- Basic order tracking
- Admin product management

Out of Scope — later release

- Loyalty points
- AI recommendations
- Marketplace sellers
- Mobile native app
- International shipping

Acceptance Criteria

How stakeholders know the deliverable is acceptable

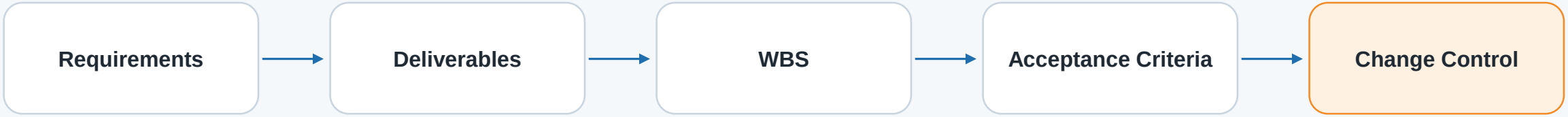
Weak criterion
“Search works well.”

Better criterion
“Search returns matching products by name/category within 2 seconds for 95% of queries.”

Good acceptance criteria are observable, testable, and agreed by stakeholders.

Scope Baseline and Traceability

Scope must be controlled after approval



Traceability helps answer: which requirement led to this work package and which test will verify it?

Interactive: Is It In Scope?

ABC Retail first release

Customer login	**Product search**	**Online payment**	**Loyalty points**
Admin product dashboard	**Arabic/English interface**	**Mobile native app**	**AI recommendations**

Students vote: In Scope, Out of Scope, or Need More Information?

What Is a Work Breakdown Structure?

A deliverable-oriented decomposition of project work



WBS answers: What work must the project complete to deliver the approved scope?

WBS Anatomy

From project to work packages



Work packages are the lowest level used for estimating, assigning responsibility, and tracking.

WBS Decomposition Rules

Keep the WBS useful and manageable

- Use the 100% rule: include all project work in the scope.
- Decompose by deliverables, not by random tasks only.
- Stop when work packages can be estimated and assigned.
- Avoid mixing deliverables with schedule dates in the WBS.
- Use clear codes: 1.0, 1.1, 1.1.1, etc.

Example WBS: ABC Retail Online Shopping System

A simplified deliverable-oriented decomposition

1.0 Project Management	2.0 Requirements	3.0 Design	4.0 Development	5.0 Testing	6.0 Deployment
1.1 Meetings	2.1 User interviews	3.1 UI design	4.1 Customer module	5.1 System test	6.1 Hosting setup
1.2 Reports	2.2 Requirements doc	3.2 DB design	4.2 Product/catalog	5.2 UAT	6.2 Training
1.3 Control	2.3 Approval	3.3 Architecture	4.3 Cart/orders	5.3 Defect fixes	6.3 Go-live

Work Package vs Activity

Do not confuse decomposition with scheduling

Work Package

- Deliverable-oriented
- Can be assigned to an owner
- Can be estimated
- Example: “Shopping cart module”

Activity

- Action-oriented
- Appears in the schedule
- Has duration and dependencies
- Example: “Develop cart API”

WBS Dictionary

Adds meaning and control to work packages

Field	Example
WBS Code	4.3.1
Work Package	Shopping Cart Module
Description	Users can add, update, remove items and view totals
Owner	Frontend/Backend developer
Acceptance Criteria	Cart total updates correctly; empty cart handled
Assumptions	Product catalog API available
Estimate basis	Similar module + expert judgment

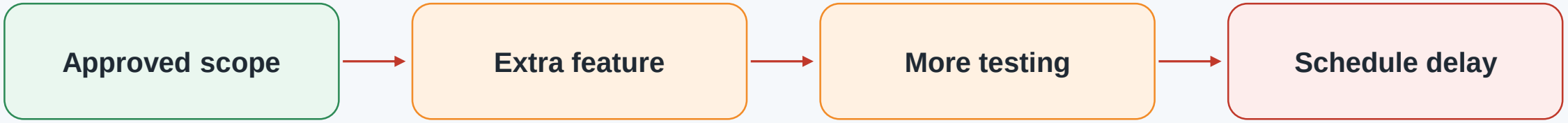
Common WBS Mistakes

Diagnose before estimating

- **Missing testing or deployment work**
- **Breaking down by team member only**
- **Putting dates inside WBS items**
- **Using vague labels such as “backend work”**
- **Mixing requirements with activities**
- **Ignoring documentation and training**

Scope Creep

Small uncontrolled changes can damage the whole plan



Scope control does not reject all changes; it ensures changes are evaluated and approved.

Week 3 Summary

From authorized project to planned scope

- Step Wise planning
- Approach selection
- Requirements
- Scope statement
- WBS
- Scope control

Next week: How much effort, time, and cost will the approved scope require?